Throughput QOS

2026-03-11-19-13-52

Objective

The objective of the QoS (Quality of Service) traffic throughput test is to measure the maximum achievable throughput of a network under specific QoS settings and conditions.By conducting this test, we aim to assess the capacity of network to handle high volumes of traffic while maintaining acceptable performance levels,ensuring that the network meets the required QoS standards and can adequately support the expected user demands.

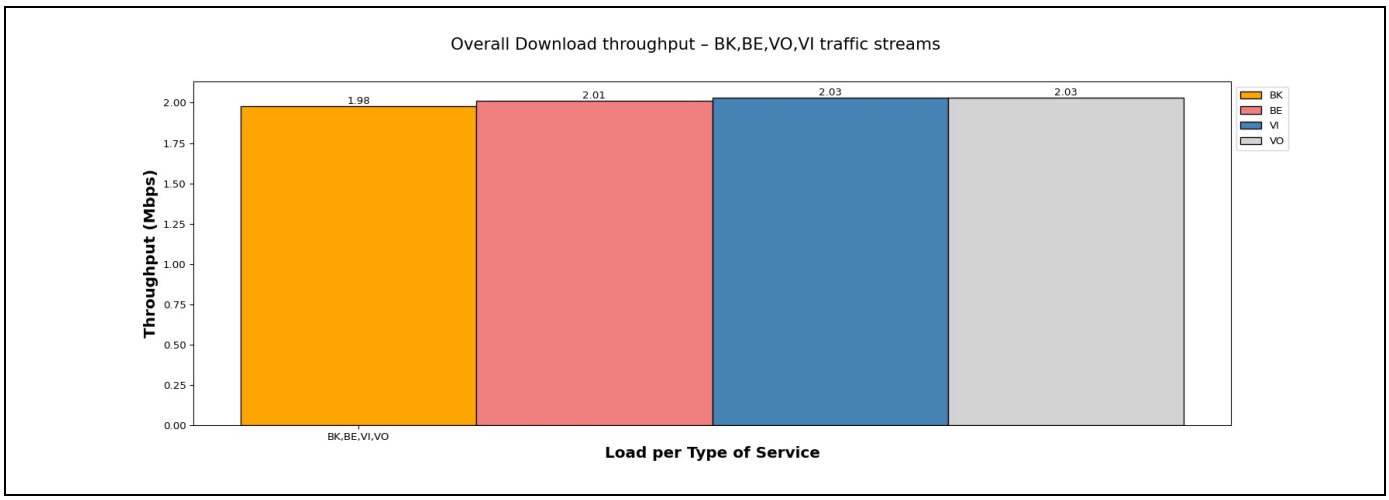
Test Configuration	Number of Stations	2
	AP Model	NETGEAR
	SSID_2.4GHz	NETGEAR_2G_wpa2
	SSID_5GHz	None
	SSID_6GHz	None
	Traffic Duration in hours	0.02
	Security_2.4GHz	wpa2
	Security_5GHz	None
	Security_6GHz	None
	Protocol	UDP
	Traffic Direction	Download
	TOS	['VO', 'VI', 'BE', 'BK']
	Per TOS Load in Mbps	1.0 Mbps

Overall Download throughput for all TOS i.e BK | BE | Video (VI) | Voice (VO)

No of Stations	Throughput for Load 1.0 Mbps-download
2	BK : 1.98, BE : 2.01, VI: 2.03, VO: 2.03

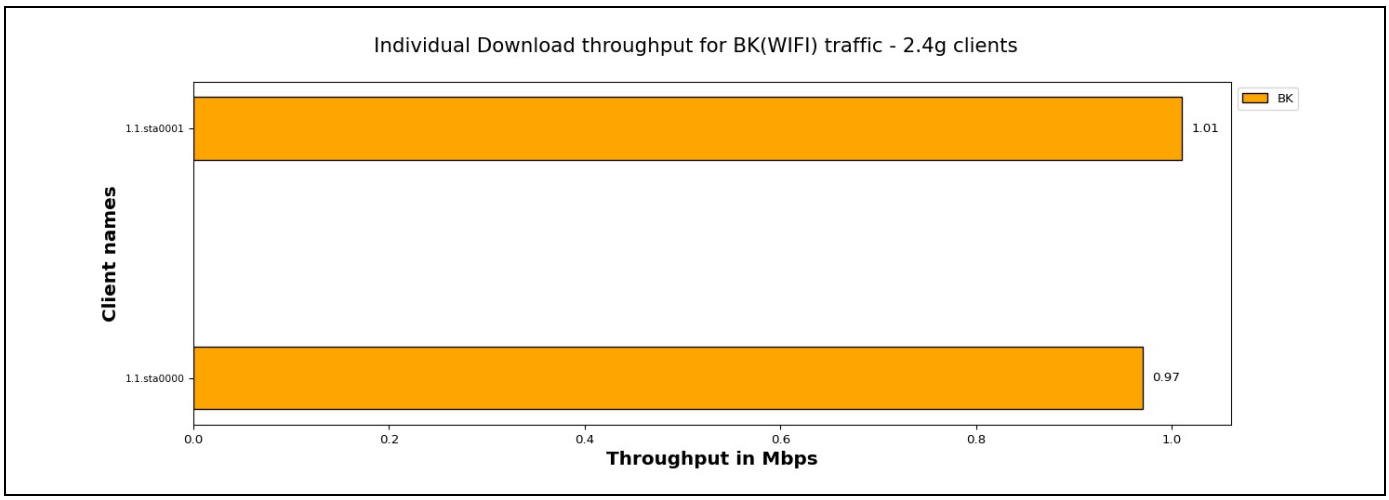
Overall Download throughput for 2 clients for 1.0 Mbps-download band with different TOS.

The below graph represents overall Download throughput for all connected stations running BK, BE, VO, VI traffic with different intended loads per station – 1.0 Mbps-download



Individual Download throughput with intended load 1.0 Mbps/station for traffic BK(WiFi).

The below graph represents individual throughput for 2 clients running BK (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

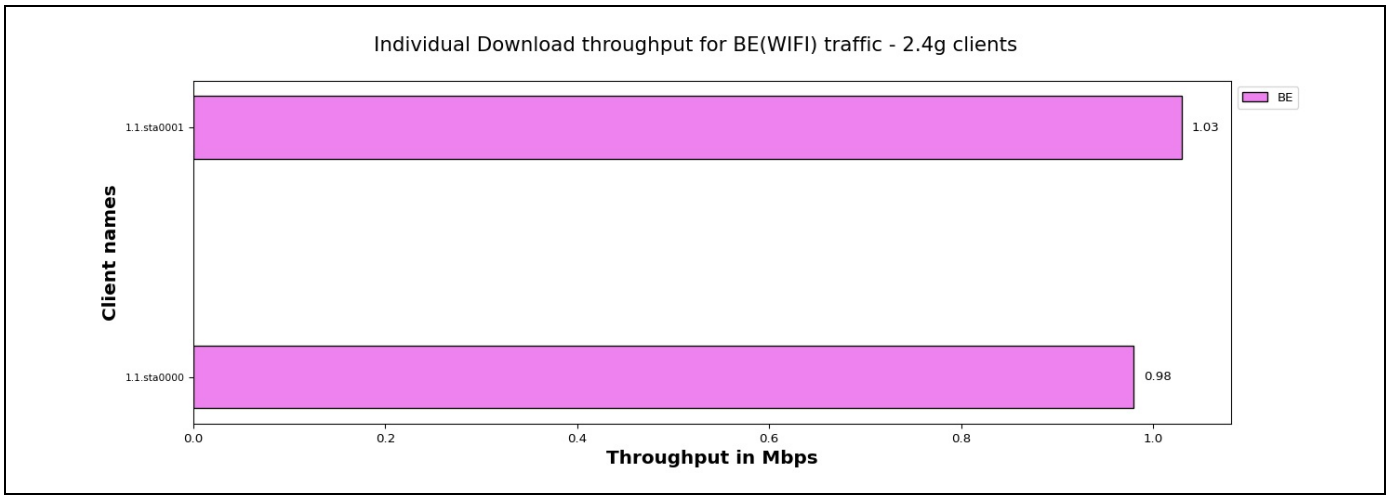


TOS : Background

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:81:46	11	NETGEAR_2G_wpa2	Background	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.97	36.527365
1.1.sta0001	04:f0:21:94:0a:46	11	NETGEAR_2G_wpa2	Background	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.01	33.980752

Individual Download throughput with intended load 1.0 Mbps/station for traffic BE(WiFi).

The below graph represents individual throughput for 2 clients running BE (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

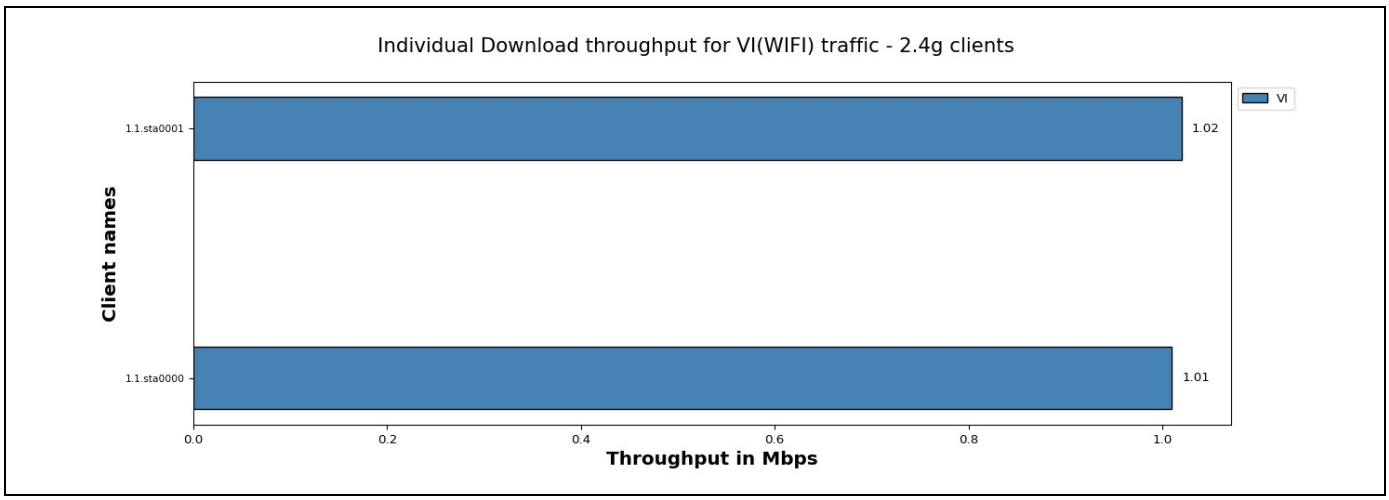


TOS : Besteffort

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:81:46	11	NETGEAR_2G_wpa2	Besteffort	Download	UDP	0.0 Mbps	1.0 Mbps	0	0.98	36.632278
1.1.sta0001	04:f0:21:94:0a:46	11	NETGEAR_2G_wpa2	Besteffort	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.03	33.922078

Individual Download throughput with intended load 1.0 Mbps/station for traffic VI(WiFi).

The below graph represents individual throughput for 2 clients running VI (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.

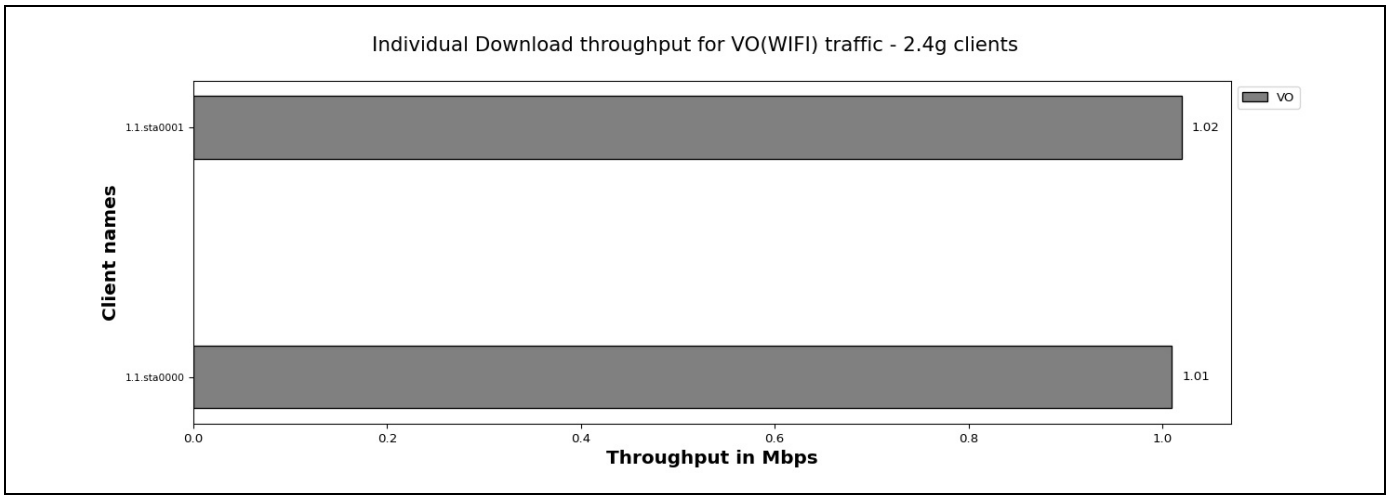


TOS : Video

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:81:46	11	NETGEAR_2G_wpa2	Video	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.01	36.577298
1.1.sta0001	04:f0:21:94:0a:46	11	NETGEAR_2G_wpa2	Video	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.02	34.298674

Individual Download throughput with intended load 1.0 Mbps/station for traffic VO(WiFi).

The below graph represents individual throughput for 2 clients running VO (WiFi) traffic. Y- axis shows “Client names“ and X-axis shows “Throughput in Mbps”.



TOS : Voice

Client Name	Mac	Channel	SSID	Type of traffic	Traffic Direction	Traffic Protocol	Offered upload rate(Mbps)	Offered download rate(Mbps)	Observed upload rate(Mbps)	Observed download rate(Mbps)	Observed Download Drop (%)
1.1.sta0000	04:f0:21:94:81:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.01	36.692374
1.1.sta0001	04:f0:21:94:0a:46	11	NETGEAR_2G_wpa2	Voice	Download	UDP	0.0 Mbps	1.0 Mbps	0	1.02	34.429900

Information	contact	support@candelatech.com